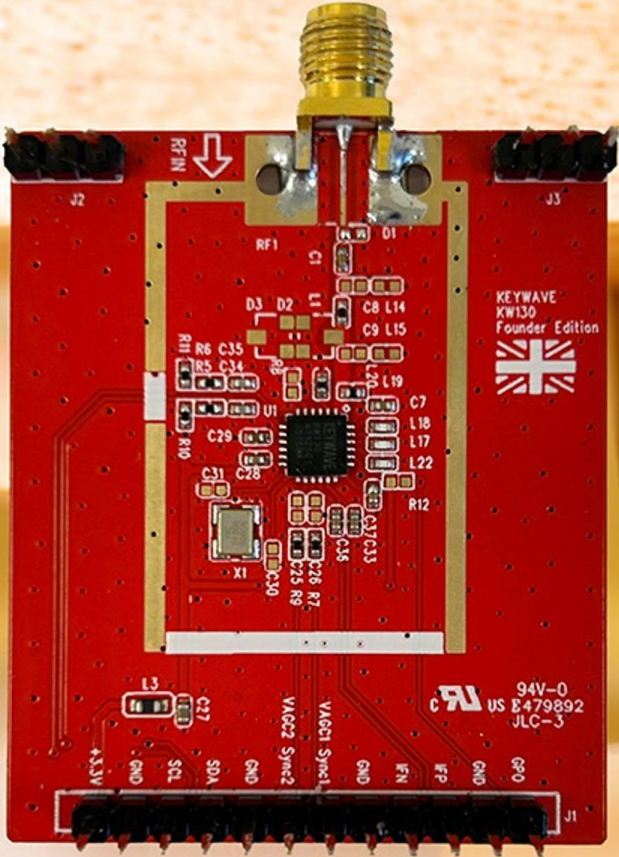


Keywave -
KW130
SDR Tuner
Programming Guide

Revision 2.0
27-Aug-2026



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Revision History			
Revision	Release Date	Author	Description
1.0	2026/05/09	Cloudio Tang	Preliminary
2.0	2026/8/27	Cloudio Tang	Rewritten. Parameter and function chapters dropped - they are in KW130_SDR.h and README.md. I2C chapter dropped - it is in the datasheet. PLL and AGC methods added.

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1. Overview

This guide covers the two things the KW130 documentation set does not state anywhere else: how the PLL frequency is calculated, and how the AGC behaves. Everything else about the part is already written down, and repeating it here would only create a second copy to keep current.

1.1 Where to look

If you need	Read
Electrical specifications, pinout, package	Datasheet
The I2C device address, transaction sequence and AC timing	Datasheet
What a register field means, its range and its reset value	Appendix A of this guide, or the Datasheet register section for the common ones
The API: every function, every set_info field, the return codes	KW130_SDR.h and README.md, both in the API package
How to implement the four host I2C functions	i2c_Sys/I2C_Sys.h in the API package
How the LO frequency is calculated	Chapter 2 of this guide
How the AGC loop behaves and what its settings do	Chapter 3 of this guide
Evaluating the part on Keywave hardware	EVK User Guide and GUI User Guide

1.2 Using the API

Most designs never need this guide. Call KW130_initial() once, then KW130_set_tuner() for every channel; the API computes the PLL, picks the per-band settings and waits for lock. KW130_standby() powers the part down, and the next KW130_set_tuner() brings it back - there is no separate wake-up call.

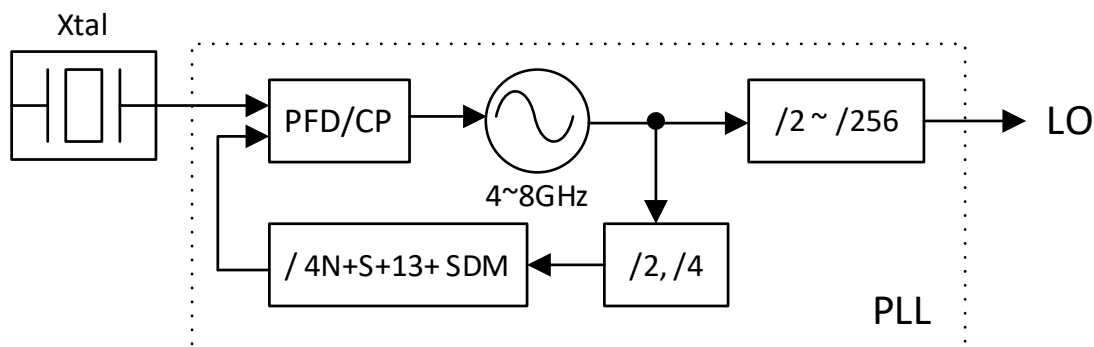
Read on if you are writing your own driver instead of using the API, if you are checking a register value you have read back, or if you are tuning AGC behaviour for a difficult signal.

2. PLL programming

The KW130 API sets the PLL for you: KW130_set_tuner() takes the RF frequency and the crystal, works out every divider, writes them and waits for lock. This note is for two readers only — anyone writing their own driver instead of using the API, and anyone checking a register value they have read back.

2.1 The divider chain

One VCO feeds two paths. The receive path divides it down to the LO the mixer needs. The feedback path divides it by the prescaler and then by the NS counter, and that is what the phase detector compares against the reference.



So the LO is set by choosing the LO divider first, which fixes the VCO frequency, and then solving the feedback path for that VCO frequency.

2.2 Step 1 — the LO divider

The VCO only runs between 4096 and 8192 MHz. Pick the smallest divider that puts the VCO inside that window:

```

for LoDiv in 2, 4, 8, 16, 32, 64, 128, 256:
    if 4096 MHz <= LO x LoDiv < 8192 MHz: use it
  
```

$$VCO = LO \times LoDiv$$

The divider is programmed by three fields, not one:

LO divider	sel_div1 R5[1]	iqgen_div2 R5[2]	sel_div R26[2:0]
/2	1	1	0
/4	0	1	0
/8	0	0	0
/16	0	0	1
/32	0	0	2
/64	0	0	3
/128	0	0	4
/256	0	0	5

Below about 32 MHz the VCO window cannot be reached even with /256, so the PLL will not lock.

2.3 Step 2 — the prescaler

The prescaler keeps the NS counter input inside its own range. It is /4 above 6000 MHz and /2 at or below it:

```

sel_vco_div24 = (VCO > 6000 MHz)      R26[3]
prescaler      = 2 x (sel_vco_div24 + 1)
  
```

2.4 Step 3 — the reference

$PLL_Ref = crystal / (div_clkp11 + 1)$ $div_clkp11 = R33[6:5], 0:/1 \ 1:/2 \ 2:/3 \ 3:/4$
KW130 normally runs with /1, so PLL_Ref is the crystal frequency. The divider is there for crystals that would otherwise put the reference above the phase detector range.

2.5 Step 4 — the loop divider

```
Nint = int( VCO / prescaler / PLL_Ref )
Nfra = VCO / prescaler / PLL_Ref - Nint
Nfra_m = int( Nfra x 2^16 )
```

$0 \leq Nfra < 1$
the 16-bit SDM value

2.6 Step 5 — Ni2c and Si2c

Nint is not written directly. It is split so that:

```
Nint = 4 x Ni2c + Si2c + 13

Ni2c = (Nint - 13) / 4      R23[7:2]
Si2c = Nint - 4 x Ni2c - 13 R23[1:0]
```

2.7 Step 6 — the SDM value

```
SDM[8:1] = Nfra_m & 0xFF      R24[7:0]
SDM[16:9] = (Nfra_m >> 8) & 0xFF R25[7:0]
pwd_sdm = (Nfra_m == 0)      R26[4] integer-N, so the modulator is off
```

2.8 Step 7 — boundary spur avoidance

A fractional-N loop produces its worst spurs when Nfra sits very close to 0, to 1, or to exactly one half. Move Nfra_m away from those three points before writing it. The largest LO error this introduces is about 47 kHz on a 24 MHz crystal.

Nfra_m from step 4	written instead	also
< 256	0	integer-N
> 65535- 256	0	Nint + 1
32640 .. 32767	32640	—
32768 .. 32895	32896	—
anything else	unchanged	—

2.9 Locking

Write every field above, then pulse rst_atune high and low. The tuner runs its own search and locks without further help. Read the lock flag only after allowing it to settle — reading too early reports a failure that is not there.

```
write R23..R33 with rst_atune = 1
write R26      with rst_atune = 0
wait 1 ms
repeat up to 10 times:
    read R4[7]          1 = locked, done
    if pw0_vco < 15: pw0_vco++ more VCO current, write R28
    wait 1 ms
```

Raising pw0_vco does not need a second rst_atune. The search falls back to a linear search and still reaches lock, only more slowly. pw0_vco starts at 10.

2.10 Worked examples

Computed with the algorithm above. LO and crystal in MHz.

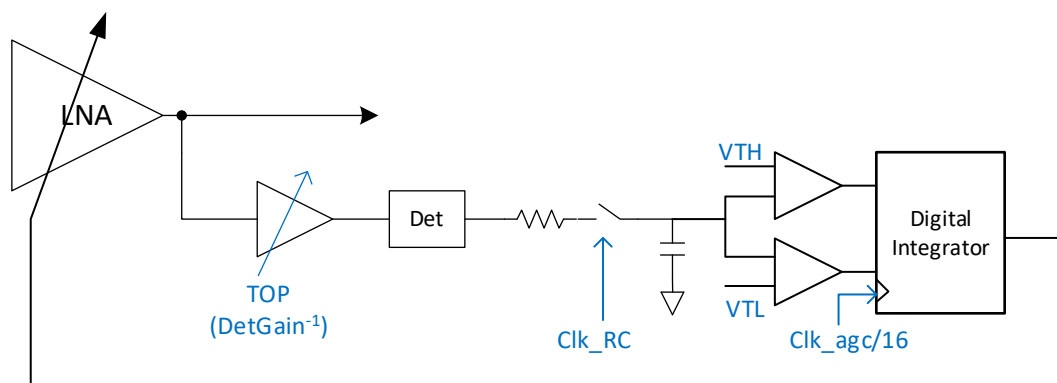
LO	Xtal	LoDiv	VCO	Pre	Nint	Nfra_m	Ni2c	Si2c	SDM[16:9]	SDM[8:1]	LO error kHz
1200	24	/4	4800	/2	100	0	21	3	0x00	0x00	+0.000
671	24	/8	5368	/2	111	54613	24	2	0xD5	0x55	-0.031
671	27	/8	5368	/2	99	26699	21	2	0x68	0x4B	-0.088

The residual LO error is the SDM quantisation step and is below 0.1 kHz across the band.

3. AGC

3.1 How the loop works

One loop is drawn below. The KW130 has two of them: an RF loop that moves the LNA and the RF buffer, and a baseband loop that moves the mixer amp and the filter. They are identical in structure and share one clock, so they always run at the same speed. Their windows and their take-off points are separate.



Reading it left to right: the block being controlled feeds a detector through a variable gain stage. That stage is the take-off point — it is the inverse of the detector gain, which is why raising DetGain lowers the TOP. The rectified level is smoothed by a switched-capacitor filter whose corner is set by Clk_RC, then compared against two thresholds. VTH says the signal is too strong, VTL says it is too weak, and a digital integrator between them drives the gain. Between the thresholds neither comparator fires and the gain stays where it is.

3.2 The knobs

Everything a host can change, and what the part comes up with:

Setting	Register	Range	Default
Window high — lna_vth, mix_vth	R34[7:4], R35[7:4]	0.4 .. 1.9 V in 0.1 V steps	1.3 V
Window low — lna_vtl, mix_vtl	R34[3:0], R35[3:0]	0.3 .. 1.8 V in 0.1 V steps	0.9 V
AGC clock, fast group	R36[1:0]	64k, 16k, 4k, 1k	4k
AGC clock, slow group	R36[4:2]	2k, 1k, 500, 256, 125, 32, 8, 2	—
Fast / slow selector	R36[5], R36[6]	see section 3.3	fast
Clk_RC, fast group	R37[1:0]	256k, 64k, 16k, 4k	16k
Clk_RC, slow group	R37[4:2]	4k, 2k, 1k, 500, 256, 64, 16, 4	—
AGC clock source	R20[7]	crystal / 2, crystal / 3	/3 above 20 MHz
TOP — LNA, RF buffer	R48[4:2], R48[7:5]	0 .. 7, 0 is the lowest TOP	4, 1
DetGain — mixer amp, filter	R49[2:0], R49[5:3]	0 .. 7, 0 = TOP max	3, 4

The LNA and RF buffer take-off points are expressed as TOP and count up. The mixer amp and filter ones are expressed as DetGain and so count the other way — a larger number is a lower take-off point. Both are the same knob in the diagram, named from opposite ends.

3.3 The fast / slow selector

Both clock groups have their own register field and both are always programmed, but only one group is in effect. A selector decides which, and it is not visible in the signal path.

Writing a slow rate into sel_clks_agc while the selector still points at the fast group changes nothing at all, and nothing reports an error. This is the most common mistake when driving the AGC by hand.

Field	Register	Meaning
-------	----------	---------

man_clk_agc_slow	R36[5]	0 = the part chooses, 1 = the host chooses
clk_agc_slow	R36[6]	0 = fast group live, 1 = slow group live

To use any slow rate, set both to 1. Clk_RC follows the same selector, so a corner is only reachable from its own group. 4 kHz is the only corner present in both.

3.4 Clock, and the rate the loop actually steps at

The value selected in R36 is the clock before the divider. The integrator receives that clock divided by 16, and that is the rate at which the gain can change. Clk_RC is not divided.

Group	Code	Clock setting	Loop steps at
fast	0	64k	4 kHz
fast	1	16k	1 kHz
fast	2	4k	250 Hz
fast	3	1k	62.5 Hz
slow	0	2k	125 Hz
slow	1	1k	62.5 Hz
slow	2	500	31.25 Hz
slow	3	256	16 Hz
slow	4	125	7.812 Hz
slow	5	32	2 Hz
slow	6	8	0.5 Hz
slow	7	2	0.125 Hz

Rates are in Hz unless marked k. So the default 4k setting steps the gain at 250 Hz. 1 kHz appears in both groups and the two are not the same — they differ in loop damping, so if that is the rate you want, try both.

4. PLL sync

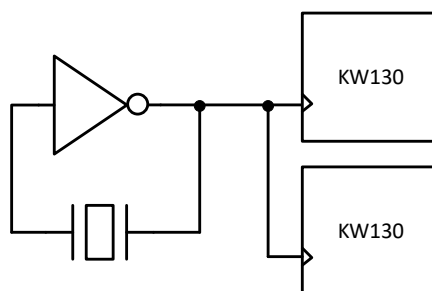
KW130 Provide two PLL/LO sync pin input for multi KW130 tuners sync the LO at the same time.

Sync1 Pin => Reset the LO Divider phase

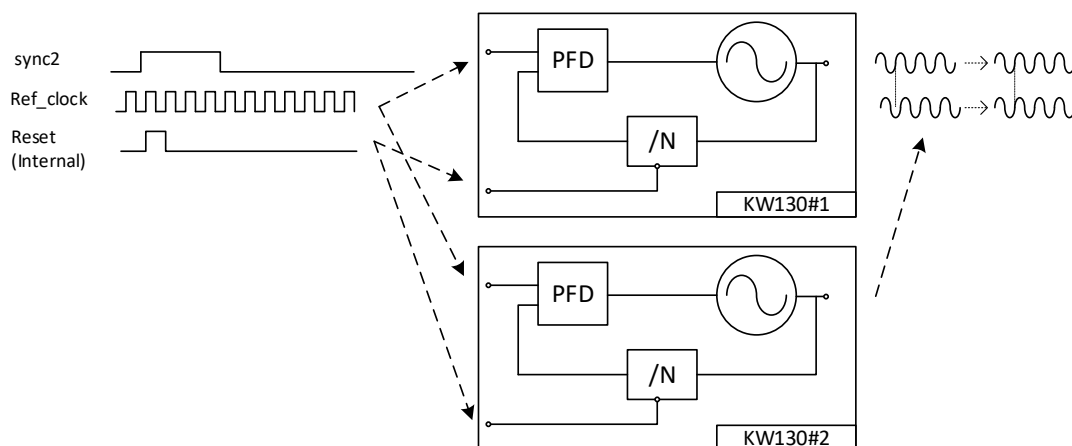
Sync2 Pin => Reset the PLL phase

Standard procedure for multi-KW130s to sync LO phase

1. Turn on the KW130 sync function: `en_sync = 1`, `R5[0]`. The KW130 API does not write this bit, so set it yourself after `KW130_initial()`.
2. Same PLL reference clock (ex. share crystal or from same off-chip crystal oscillator).

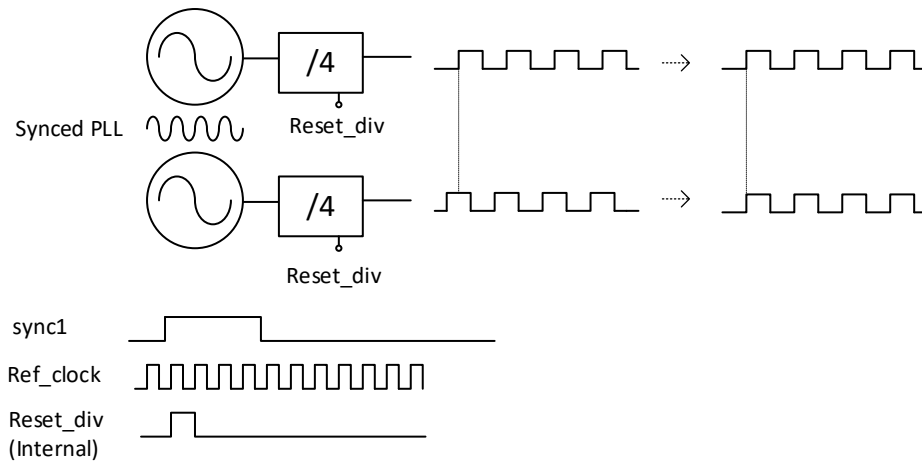


3. Set multi-PLLs to the same LO frequency.
4. After PLLs locks (Lock flag =1), send sync2 pulse to KW130 to sync PLL phase
 - a. Pulse width > reference period (ex. > 40ns for 25MHz reference)
 - b. Pulse can be once or periodic $N \cdot T_{ref}$ (N suggest > 65536)
 - c. Stop sync pulse in normal operation



(Note: PLL must lock before input sync. Let PLL frequency stable before sync phase)

5. After PLLs sync2, multi-KW130 may still out of phase due to divider $/N$ have N output phase options. Sync1 is used to generate a reset pulse to sync divider output.



(Note: Sync1 function is not perfect right now, it may have 2~3 output phases results)

Appendix A - Register Map

v1.0. All 59 control registers, R5 to R63.

R0–R4 are read-only status registers and are documented separately. R5–R63 are control registers. *Init* is the value to write during initialisation.

Reserved — write the *Init* value shown and nothing else.

Reserved — API controlled — the KW130 API sets this bit for you. Do not write it directly.

R5 — 0x05

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	en_rfdet_ibst	0	en_rfdet_ibst	—
3	lovcm_up	0	lovcm_up	—
2	lqgen_div2	0	IQGen div	—
1	sel_div1	0	VCO div1	—
0	en_sync	0	en sync function	—

R6 — 0x06

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3:2	mixeramp_swC	0	mixeramp BW	0 = 0:normal 1 = 1 2 = 2 3 = 3:widest
1:0	BWS	0	filter coarse BW	0 = normal 2 = 12MHz 3 = 16/20MHz

R7 — 0x07

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R8 — 0x08

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—

Bits	Field	Init	Description	Values
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R9 — 0x09

Init value: 0xEC

Bits	Field	Init	Description	Values
7	en_pll	1	enable block power	0 = OFF 1 = ON
6	en_mixer	1	enable block power	0 = OFF 1 = ON
5	en_rfbuf	1	enable block power	0 = OFF 1 = ON
4	Reserved	0	Must be written 0.	—
3	en_lna	1	enable block power	0 = OFF 1 = ON
2	en_BG	1	enable block power	0 = OFF 1 = ON
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R10 — 0x0A

Init value: 0x9C

Bits	Field	Init	Description	Values
7	en_xtal_det	1	enable block power	0 = OFF 1 = ON
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	1	Must be written 1.	—
3	en_lna_rf_det	1	enable block power	0 = OFF 1 = ON
2	en_BB	1	enable block power	0 = OFF 1 = ON
1	pwd_xtalldo	0	power down block power	0 = ON 1 = OFF
0	pwd_xtal	0	power down block power	0 = ON 1 = OFF

R11 — 0x0B

Init value: 0x40

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	Reserved	1	Must be written 1.	—
5	Reserved	0	Must be written 0.	—
4	rst_agc_i2c	0	Reset block	0 = No_Rst 1 = Reset
3	Reserved	0	Reserved — API controlled	—
2	Reserved	0	Reserved — API controlled	—
1	Reserved	0	Reserved — API controlled	—
0	pwd_agc	0	power down block power	0 = ON 1 = OFF

R12 — 0x0C

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	Reserved	0	Reserved — API controlled	—
5	Reserved	0	Reserved — API controlled	—
4	Reserved	0	Reserved — API controlled	—
3	Reserved	0	Reserved — API controlled	—
2	Reserved	0	Reserved — API controlled	—
1	Reserved	0	Reserved — API controlled	—
0	Reserved	0	Reserved — API controlled	—

R13 — 0x0D

Init value: 0x20

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	Reserved	0	Reserved — API controlled	—
5	en_code_mode	1	enable vga code mode	0 = OFF 1 = ON
4	Reserved	0	Reserved — API controlled	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Reserved — API controlled	—
1	Reserved	0	Reserved — API controlled	—
0	Reserved	0	Reserved — API controlled	—

R14 — 0x0E

Init value: 0xA8

Bits	Field	Init	Description	Values
7	Reserved	1	Must be written 1.	—
6	Reserved	0	Reserved — API controlled	—
5	Reserved	1	Reserved — API controlled	—
4	qlimit	0	TF Q factor degeneration	0 = 900 1 = 1350
3	en_qlimit	1	enable UHF TF Q factor degeneration	0 = OFF 1 = ON
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	man_lna_gain	0	manual LNA gain	0 = auto 1 = manual

R15 — 0x0F

Init value: 0x15

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5:4	pw3_lna	1	LNA s2d current setting	0 (min) .. 3 (max)
3:2	pw1_lna	1	LNA high pin mode current setting	0 (min) .. 3 (max)
1:0	pw0_lna	1	LNA low pin mode current setting	0 (min) .. 3 (max)

R16 — 0x10

Init value: 0x24

Bits	Field	Init	Description	Values
7:6	polyrf_mux	0	polyrf frequency band selection	0 = High 1 = Middle 2 = Low 3 = Highest
5:4	Band	2	LNA frequency band selection	0 = UHF 1 = HF 2 = MF 3 = LF
3	Reserved	0	Must be written 0.	—
2	Reserved	1	Must be written 1.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R17 — 0x11

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6:0	TF	0	TF Cap MSB	0 .. 127

R18 — 0x12

Init value: 0x28

Bits	Field	Init	Description	Values
7:6	rf_lpf	0	Rfbuf LPF	0 = no LPF 1 = 250MHz 2 = 170MHz 3 = 130MHz
5	Reserved	1	Must be written 1.	—
4	man_rf_gain	0	manual mode rfbuf gain	0 = auto 1 = manual
3:0	rf_gain	8	rfbuf gain setting	0 .. 15

R19 — 0x13

Init value: 0x29

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	1	Must be written 1.	—
4:3	pw1_rfbuf	1	rfbuf current setting	0 (min) .. 3 (max)
2	pw0_rfbuf	0	rfbuf degeneration R setting	0 = high 1 = low
1	Reserved	0	Must be written 0.	—
0	Reserved	1	Must be written 1.	—

R20 — 0x14

Init value: 0xA2

Bits	Field	Init	Description	Values
7	mod23_clkagc	1	clk agc /2/3 from xtal	0 = /2 1 = /3
6	rfvcm	0	mixer rf vcm setting	0 = 2.55V 1 = 2.7V

Bits	Field	Init	Description	Values
5:4	LO_vdd	2	LO buffer vdd setting	0 = 1.5V 1 = 1.6V 2 = 1.7V 3 = 1.8V
3	Reserved	0	Reserved — API controlled	—
2	Reserved	0	Reserved — API controlled	—
1:0	pw_mix	2	mixer current setting	0 (min) .. 3 (max)

R21 — 0x15

Init value: 0x10

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4:0	lna_gain	16	LNA gain setting = $16 * \text{lna_gain}[4] + \text{lna_gain}[3:0]$	0 .. 31

R22 — 0x16

Init value: 0x43

Bits	Field	Init	Description	Values
7:6	sel_atune (pll)	1	pll autotune speed	0 = 128k 1 = 64k 2 = 32k 3 = 16k
5	pw0_xtalldo	0	Xtal ldo max current	0 = 1mA 1 = 1.5mA
4	Reserved	0	Must be written 0.	—
3:0	pw0_xtal	3	Xtal current setting	0 = 0u 1 = 40u 2 = 80u 3 = 120u ... 15 = 600u

R23 — 0x17

Init value: 0x62

Bits	Field	Init	Description	Values
7:2	Ni2c	24	NS Divider = $4N+S+13$	0 .. 63
1:0	Si2c	2	NS Divider = $4N+S+13$	0 .. 3

R24 — 0x18

Init value: 0x55

Bits	Field	Init	Description	Values
7:0	SDM_IN	85	SDM num LSB	0 .. 255

R25 — 0x19

Init value: 0xD5

Bits	Field	Init	Description	Values
7:0	SDM_IN	213	SDM num MSB	0 .. 255

R26 — 0x1A

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	rst_atune	0	reset autotune	0 = No Rst 1 = Reset
5	search_mode	0	autotune search mode	0 = binary 1 = linear
4	pwd_sdm	0	SDM power	0 = ON 1 = OFF
3	sel_vco_div24	0	VCO to Nsdivider Div Num	0 = /2 1 = /4
2:0	sel_div	0	VCO to Mixer Divider Num	0 = /8 1 = /16 2 = /32 3 = /64 4 = /128 5 = off 6 = off 7 = off

R27 — 0x1B

Init value: 0x80

Bits	Field	Init	Description	Values
7:6	pw0_cp	2	charge pump current	0 = 800u 1 = 700u 2 = 600u 3 = 500u
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1:0	bit_mode	0	PLL autotune search bit	0 = 8bit 1 = 7bit 2 = 6bit 3 = 5bit

R28 — 0x1C

Init value: 0xCA

Bits	Field	Init	Description	Values
7	Reserved	1	Must be written 1.	—
6	Reserved	1	Must be written 1.	—
5	iqgen_swi	0	iqgen current	0 = 80u 1 = 40u
4	iqgen_swl	0	iqgen latch load	0 = 1k 1 = 2k
3:0	pw0_vco	10	VCO current	0 = 0-off 1 = 1-min 2 = 2 3 = 3 ... 15 = 15-max

R29 — 0x1D

Init value: 0x07

Bits	Field	Init	Description	Values
7	pw0_dldo	0	Nsdiv LDO max current	0 = 4mA 1 = 5mA
6	sw_vol_aldo	0	Divider LDO voltage	0 = 2V 1 = 2.2V
5	deadzone	0	PFD dead zone	0 = 0.5ns 1 = 1.3ns
4	Reserved	0	Must be written 0.	—
3	pw0_cpldo	0	CP LDO max current	0 = 3mA 1 = 4.5mA
2	Reserved	1	Must be written 1.	—
1	en_sah_lf	1	SAH Loop Filter	0 = OFF 1 = ON
0	en_pfd_offset	1	PFD offset mode	0 = OFF 1 = ON

R30 — 0x1E

Init value: 0x82

Bits	Field	Init	Description	Values
7	en_autopw	1	vco auto power mode	0 = manual 1 = auto
6	pw0_vcdiv2	0	current setting	0 = 80u 1 = 120u
5	pw0_vcobuf	0	current setting	0 = 80u 1 = 120u
4	pw0_iqgen	0	current setting	0 = 80u 1 = 120u
3	pw0_hfd	0	current setting	0 = 80u 1 = 120u
2	pw0_divbuf	0	current setting	0 = 80u 1 = 120u
1	Reserved	1	Must be written 1.	—
0	man_vco_th	0	manual autotune threshold	0 = auto 1 = manual

R31 — 0x1F

Init value: 0x00

Bits	Field	Init	Description	Values
7:4	sel_vco	0	manual vcoL vcoH	0 .. 15
3:0	(not documented)	0	No entry for these bits in the source spreadsheet.	—

R32 — 0x20

Init value: 0x3F

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	1	Reserved — API controlled	—
4	Reserved	1	Must be written 1.	—
3	en_dldo	1	enable block power	0 = OFF 1 = ON
2	en_aldo	1	enable block power	0 = OFF 1 = ON

Bits	Field	Init	Description	Values
1	en_vco_ldo/en_vco	1	enable vco/vco_ldo power	0 = OFF 1 = ON
0	en_cp	1	enable cp power	0 = OFF 1 = ON

R33 — 0x21

Init value: 0x18

Bits	Field	Init	Description	Values
7	pol_clkpll	0	XTAL to PLL polarity	0 = neg 1 = pos
6:5	div_clkpll	0	XTAL to PLL divider	0 = /1 1 = /2 2 = /3 3 = /4
4	sw_detref	1	XTAL AAC det reference	0 = 0.6Vpp 1 = 0.4Vpp
3	Reserved	1	Must be written 1.	—
2	sel_xixo	0	sel xtal from xi or xo	0 = xo 1 = xi
1:0	dampR_xo	0	xo damping resistor	0 = 0R 1 = 250R 2 = 500R 3 = 750R

R34 — 0x22

Init value: 0x96

Bits	Field	Init	Description	Values
7:4	lna_vth	9	AGC threshold high	0 = 0.4 1 = 0.5 2 = 0.6 3 = 0.7 ... 15 = 1.9
3:0	lna_vtl	6	AGC threshold low	0 = 0.3 1 = 0.4 2 = 0.5 3 = 0.6 ... 15 = 1.8

R35 — 0x23

Init value: 0x96

Bits	Field	Init	Description	Values
7:4	mix_vth	9	AGC threshold high	0 = 0.4 1 = 0.5 2 = 0.6 3 = 0.7 ... 15 = 1.9
3:0	mix_vtl	6	AGC threshold low	0 = 0.3 1 = 0.4 2 = 0.5 3 = 0.6 ... 15 = 1.8

R36 — 0x24

Init value: 0x32

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	clk_agc_slow	0	slow control at manual slow mode	0 = fast 1 = slow
5	man_clk_agc_slow	1	Manual slow mode	0 = auto 1 = manual
4:2	sel_clks_agc	4	AGC clock speed at slow mode (enter=clkf/16)	0 = 2k 1 = 1k 2 = 500 3 = 256 4 = 125 5 = 32 6 = 8 7 = 2
1:0	sel_clkf_agc	2	AGC clock speed at fast mode (enter=clkf/16)	0 = 64k 1 = 16k 2 = 4k 3 = 1k

R37 — 0x25

Init value: 0x12

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4:2	sel_clks_disc	4	peak detector clock speed at slow mode	0 = 4k 1 = 2k 2 = 1k 3 = 500 4 = 256 5 = 64 6 = 16 7 = 4
1:0	sel_clkf_disc	2	peak detector clock speed at fast mode	0 = 256k 1 = 64k 2 = 16k 3 = 4k

R38 — 0x26

Init value: 0x87

Bits	Field	Init	Description	Values
7	Reserved	1	Must be written 1.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	1	Must be written 1.	—
1	Reserved	1	Must be written 1.	—
0	Reserved	1	Must be written 1.	—

R39 — 0x27

Init value: 0x0A

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3:2	Reserved	2	Must be written 2.	—
1:0	Reserved	2	Must be written 2.	—

R40 — 0x28

Init value: 0xC9

Bits	Field	Init	Description	Values
7:6	man_poly_gain	3	polyphase filter buffer gain setting	0 = 0 dB 1 = 1dB 2 = 2dB 3 = 3dB
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	1	Must be written 1.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	1	Must be written 1.	—

R41 — 0x29

Init value: 0x94

Bits	Field	Init	Description	Values
7:5	IF_CH_BW	4	IF channel bandwidth setting	0 = 9MHz 4 = 8MHz 6 = 7MHz 7 = 6MHz
4:0	LPF_cor	20	LPF filter bandwidth setting	0:narrowest .. 31:widest

R42 — 0x2A

Init value: 0x0B

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3:0	HPF_cor	11	HPF corner	0 = 2.72M 1 = 2.56M 2 = 2.40M 3 = 2.24M ... 15 = 0.32M

R43 — 0x2B

Init value: 0x00

Bits	Field	Init	Description	Values
7:4	yi_phs_i	0	IMR set when mixeramp_agc <8	0 .. 15
3:0	xi_gain_i	0	IMR set when mixeramp_agc <8	0 .. 15

R44 — 0x2C

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	img_Rev	0	wanted image signal reverse	0 = normal 1 = rev
5	Reserved	0	Reserved — API controlled	—
4	Reserved	0	Reserved — API controlled	—
3:2	mpcr_gain	0	MPCR Gain	0 = 2dB 1 = 3.5dB 2 = 5.0dB 3 = 6.5dB
1	yi_sign_i	0	IMR set when mixeramp_agc <8	0 = POS 1 = NEG
0	xi_sign_i	0	IMR set when mixeramp_agc <8	0 = POS 1 = NEG

R45 — 0x2D

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	Reserved	0	Reserved — API controlled	—
5	Reserved	0	Reserved — API controlled	—
4	Reserved	0	Reserved — API controlled	—
3	Reserved	0	Reserved — API controlled	—
2	Reserved	0	Reserved — API controlled	—
1	Reserved	0	Reserved — API controlled	—
0	Reserved	0	Reserved — API controlled	—

R46 — 0x2E

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Reserved — API controlled	—
6	Reserved	0	Reserved — API controlled	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Reserved — API controlled	—
0	Reserved	0	Reserved — API controlled	—

R47 — 0x2F

Init value: 0x08

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3:0	vga_code_i	8	vga code mode gain setting	0 = 0~15

R48 — 0x30

Init value: 0x32

Bits	Field	Init	Description	Values
7:5	top_rfdet	1	RFBuf TOP	0 .. 7
4:2	top_lna	4	LNA TOP	0 .. 7
1	Reserved	1	Reserved — API controlled	—
0	Reserved	0	Reserved — API controlled	—

R49 — 0x31

Init value: 0x23

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5:3	detgain_filt	4	TOP filter	0 = TOP max 1 = TOP max-1 2 = TOP max-2 3 = TOP max-3 4 = TOP max-4 5 = TOP max-5 6 = TOP max-6 7 = TOP max-7
2:0	detgain_miamp	3	TOP mixeramp	0 = TOP max 1 = TOP max-1 2 = TOP max-2 3 = TOP max-3 4 = TOP max-4 5 = TOP max-5 6 = TOP max-6 7 = TOP max-7

R50 — 0x32

Init value: 0x07

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3:0	detgain_BB	7	TOP_ BB	0 = TOP max 1 = TOP max-1 2 = TOP max-2 3 = TOP max-3 ... 15 = TOP max-15

R51 — 0x33

Init value: 0x2A

Bits	Field	Init	Description	Values
7:6	pw_2nd_LPF	0	LPF 2 current setting	0 (min) .. 3 (max)
5:4	pw_1th_LPF	2	LPF 1 current setting	0 (min) .. 3 (max)
3:2	pw_polyphase_filt	2	polyphase current setting	0 (min) .. 3 (max)
1:0	pw_mixeramp	2	mixeramp current setting	0 (min) .. 3 (max)

R52 — 0x34

Init value: 0x80

Bits	Field	Init	Description	Values
7:4	BBamp_G	8	mixeramp gain setting	0 = 0~15
3:2	pw_vga	0	vga current setting	0 (min) .. 3 (max)
1:0	pw_HPF	0	HPF current setting	0 (min) .. 3 (max)

R53 — 0x35

Init value: 0x28

Bits	Field	Init	Description	Values
7:6	mixeramp_glimit	0	mixeramp gain limit	0 = 11 1 = 10 2 = 9 3 = 8
5	Reserved	1	Must be written 0.	—
4	man_filt_G	0	Manual mode filter gain	0 = auto 1 = manual
3:1	filt_G	4	filter gain at manual mode	0 .. 7
0	man_BBamp_G	0	Manual mode mixeramp gain	0 = auto 1 = manual

R54 — 0x36

Init value: 0x7F

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	1	Must be written 1.	—
5	Reserved	1	Must be written 1.	—
4	Reserved	1	Must be written 1.	—
3	en_vga	1	enable vga	0 = OFF 1 = ON
2	en_HPF	1	enable HPF power	0 = OFF 1 = ON
1	en_1th_2nd_LPF	1	enable LPF 1 and 2 power	0 = OFF 1 = ON
0	en_mixeramp_polyf	1	enable mixeramp power,enable polyphase power	0 = OFF 1 = ON

R55 — 0x37

Init value: 0x8E

Bits	Field	Init	Description	Values
7	Reserved	1	Must be written 1.	—
6:5	vga_EQ	0	vga equalization cap	0 .. 3
4	Reserved	0	Must be written 0.	—
3	Reserved	1	Must be written 1.	—
2	Reserved	1	Must be written 1.	—
1	Reserved	1	Must be written 1.	—
0	Reserved	0	Must be written 0.	—

R56 — 0x38

Init value: 0x3F

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6:5	eq_2nd_LPF	1	LPF equalization 2	0 .. 3
4:3	eq_1th_LPF	3	LPF equalization 1	0 .. 3

Bits	Field	Init	Description	Values
2	Reserved	1	Must be written 1.	—
1	Reserved	1	Must be written 1.	—
0	Reserved	1	Must be written 1.	—

R57 — 0x39

Init value: 0x30

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6:4	vga_att	3	vga output attenuation	0 = 0~7
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R58 — 0x3A

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R59 — 0x3B

Init value: 0xC0

Bits	Field	Init	Description	Values
7	Reserved	1	Must be written 1.	—
6	Reserved	1	Must be written 1.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R60 — 0x3C

Init value: 0xD7

Bits	Field	Init	Description	Values
7	Reserved	1	Reserved — API controlled	—
6	Reserved	1	Reserved — API controlled	—
5	Reserved	0	Reserved — API controlled	—
4	Reserved	1	Reserved — API controlled	—
3	Reserved	0	Reserved — API controlled	—
2	Reserved	1	Reserved — API controlled	—
1	Reserved	1	Reserved — API controlled	—
0	Reserved	1	Reserved — API controlled	—

R61 — 0x3D

Init value: 0x00

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5	Reserved	0	Must be written 0.	—
4	Reserved	0	Must be written 0.	—
3	Reserved	0	Must be written 0.	—
2	Reserved	0	Must be written 0.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R62 — 0x3E

Init value: 0x1C

Bits	Field	Init	Description	Values
7	Reserved	0	Must be written 0.	—
6	Reserved	0	Must be written 0.	—
5:4	mode_gpio	1	GPIO pin mode	—
3	Reserved	1	Must be written 1.	—
2	Reserved	1	Must be written 1.	—
1	Reserved	0	Must be written 0.	—
0	Reserved	0	Must be written 0.	—

R63 — 0x3F

Init value: 0x70

Bits	Field	Init	Description	Values
7	gpo_in	0	GPIO output mode voltage	0 = Output = 0 1 = Output = 1
6	Reserved	1	Must be written 1.	—
5	Reserved	1	Must be written 1.	—
4:0	cap_x	16	Xtal cap, unit 1pF	0 .. 31

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